

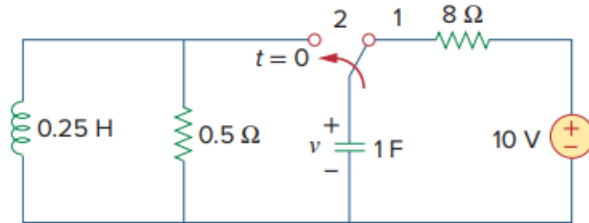
Problem

In the circuit of Fig. 8.104, the switch has been in position 1 for a long time but moved to position 2 at $t = 0$. Find:

(a) $v(0^+)$, $dv(0^+)/dt$,

(b) $v(t)$ for $t \geq 0$.

Figure 8.104:



Step-by-step solution

Step 1 of 11

Refer to Figure 8.104 in the text book.

(a)

At $t < 0$, the capacitor acts as open circuit.

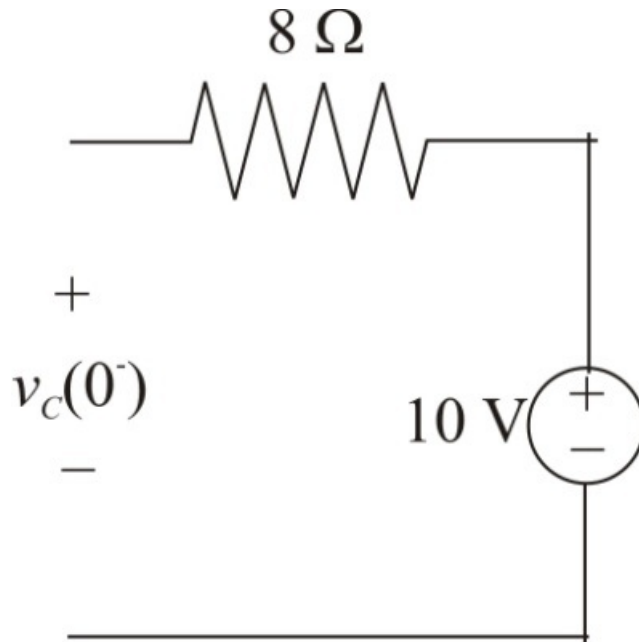
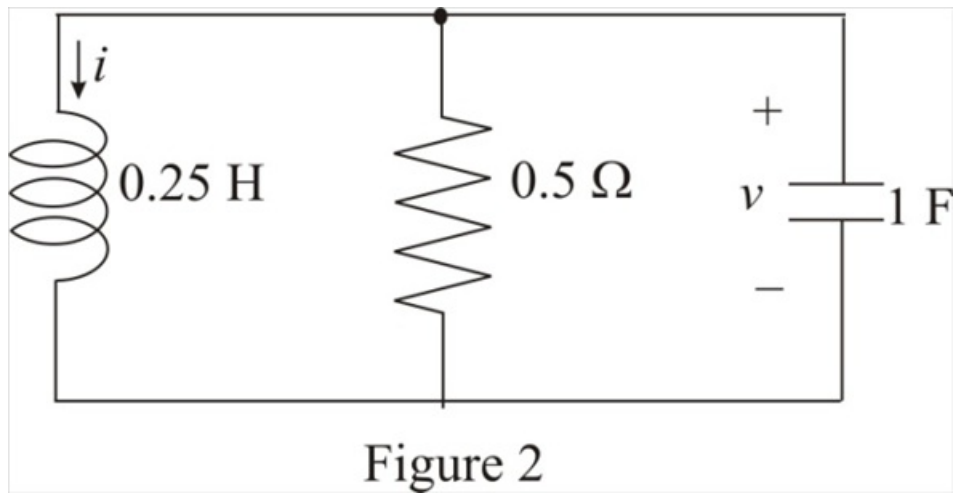


Figure 1

Therefore, the value of the voltage $v(0^-)$ is 10 V.

Step 2 of 11

Draw the circuit diagram at $t < 0$.



Step 3 of 11

The voltage across the capacitor will not change instantaneously. Hence, at $t > 0$ the value of the voltage across the capacitor $v(0^+)$ is 10 V.

Notice that the voltage across the capacitor is continuous. The value at $t = 0$ is $v(0) = 10 \text{ V}$. Also notice that the value of the current $i(0)$ is 0 A since inductor is connected at $t = 0$.

Step 4 of 11

Apply Kirchhoff's Current Law at common node.

$$1 \frac{dv}{dt} + \frac{v}{0.5} + i = 0$$

$$\frac{dv}{dt} + 2v + i = 0$$

$$\frac{dv}{dt} = -2v - i$$

Calculate the value of $\frac{dv}{dt}$ at $t = 0$.

$$\frac{dv(0)}{dt} = -2v(0) - i(0)$$

Substitute 10 V for $v(0)$ and 0 A for $i(0)$.

$$\begin{aligned} \frac{dv(0)}{dt} &= -2(10) - 0 \\ &= -20 \text{ V/s} \end{aligned}$$

Therefore, the value of $\frac{dv(0)}{dt}$ is -20 V/s.

Step 5 of 11

(b)

Apply Kirchhoff's Current Law at common node.

$$1 \frac{dv}{dt} + \frac{v}{0.5} + \frac{1}{0.25} \int v \, dt = 0$$

$$\frac{dv}{dt} + 2v + 4 \int v \, dt = 0$$

Apply derivative with respect to t .

$$\frac{d^2v}{dt^2} + 2 \frac{dv}{dt} + 4v = 0$$

Treat $\frac{d}{dt}$ as s .

$$s^2v + 2sv + 4v = 0$$

$$(s^2 + 2s + 4)v = 0$$

Step 6 of 11

Consider the characteristic equation of the circuit.

$$s^2 + 2s + 4 = 0$$

$$(s + 1 + j1.732)(s + 1 - j1.732) = 0$$

The roots of the characteristic equation are $s = -1 \pm j1.732$.

The circuit exhibits under-damped response since the roots of the characteristic equation are complex.

Compare the roots to $s = -\alpha \pm j\omega_d$.

The values of α and ω_d are 1 Np/s and 1.732 rad/s respectively.

Step 7 of 11

Draw the circuit for $t = \infty$.

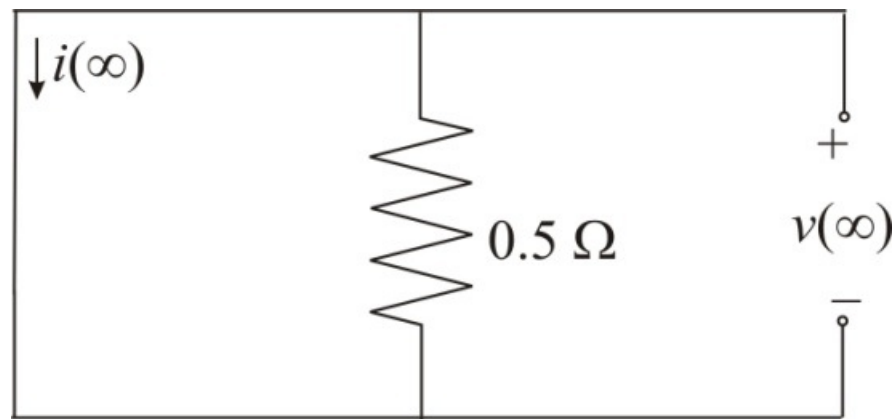


Figure 3

The value of the current $i(\infty)$ is 0 A and the value of the voltage $v(\infty)$ 0 V.

Step 8 of 11

Write the general solution for the voltage $v(t)$ for under-damped condition.

$$v(t) = v(\infty) + (A_1 \cos \omega_d t + A_2 \sin \omega_d t) e^{-\alpha t}$$

Substitute 0 V for $v(\infty)$, 1.732 rad/s for ω_d and 1 Np/s for α .

$$v(t) = 0 + (A_1 \cos 1.732t + A_2 \sin 1.732t) e^{-t} \quad v(t) = (A_1 \cos 1.732t + A_2 \sin 1.732t) e^{-t} \dots\dots (1)$$

Step 9 of 11

Recall equation-1.

$$v(t) = (A_1 \cos 1.732t + A_2 \sin 1.732t) e^{-t}$$

Substitute 0 for t .

$$\begin{aligned} v(0) &= (A_1 \cos 1.732(0) + A_2 \sin 1.732(0)) e^{-(0)} \\ &= (A_1 \cos(0) + A_2 \sin(0)) e^{(0)} \\ &= (A_1(1) + A_2(0))(1) \\ &= A_1 + 0 \end{aligned}$$

Substitute 10 V for $v(0)$.

$$A_1 = 10$$

Therefore, the value of the constant A_1 is 10.

Step 10 of 11

Recall equation-1.

$$v(t) = (A_1 \cos 1.732t + A_2 \sin 1.732t) e^{-t}$$

Apply derivative with respect to t .

$$\frac{dv(t)}{dt} = \left\{ (A_1 \cos 1.732t + A_2 \sin 1.732t)(-1)e^{-t} + e^{-t}(-1.732A_1 \sin 1.732t + 1.732A_2 \cos 1.732t) \right\}$$

Substitute 0 for t .

$$\begin{aligned} \frac{dv(0)}{dt} &= \left\{ (A_1 \cos 1.732(0) + A_2 \sin 1.732(0))(-1)e^{-(0)} + e^{-(0)}(-1.732A_1 \sin 1.732(0) + 1.732A_2 \cos 1.732(0)) \right\} \\ &= (A_1 \cos(0) + A_2 \sin(0))(-1)e^{(0)} + e^{(0)}(-1.732A_1 \sin(0) + 1.732A_2 \cos(0)) \\ &= (A_1(1) + A_2(0))(-1)(1) + (1)(-1.732A_1(0) + 1.732A_2(1)) \\ &= -A_1 + 1.732A_2 \end{aligned}$$

Substitute -20 V/s for $\frac{dv(0)}{dt}$ and 10 for A_1 .

$$-20 = (-1)(10) + 1.732A_2$$

$$-10 = 1.732A_2$$

$$A_2 = -\frac{10}{1.732}$$

$$A_2 = -5.77$$

Therefore, the value of the constant A_2 is -5.77.

Step 11 of 11

Recall equation (1).

$$v(t) = (A_1 \cos 1.732t + A_2 \sin 1.732t)e^{-t}$$

Substitute -5.77 for A_2 and 10 for A_1 .

$$v(t) = (10 \cos 1.732t - 5.77 \sin 1.732t)e^{-t} \text{ V}$$

Therefore, the value of the voltage $v(t)$ is $(10 \cos 1.732t - 5.77 \sin 1.732t)e^{-t} \text{ V}$.